

DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 6 (AT3)

Case Study

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1. Case Study 1 – Retail Sales Performance Dashboard Scenario

i.) Suitable Visualization for Each Requirement

Requirement	Best Visualization
Regional revenue comparison	Bar Plot
Monthly sales trend	Stacked Bar Chart
Product category contribution	Histogram
Advertising expenditure vs Sales	Scatter Plot
Regional sales distribution	Density Plot

ii.) R Program

Dataset

```
# Retail Sales Dataset

region <- c("North", "South", "East", "West", "Central")

revenue <- c(950, 820, 1100, 760, 900)

sales <- c(88, 76, 102, 70, 84)

advertising <- c(45, 38, 55, 32, 42)

category <- c("Electronics", "Clothing", "Furniture",
```

```

        "Groceries", "Accessories")

monthly <- matrix(c(
80,85,90,95,100,
78,82,88,91,96,
85,90,95,100,105,
82,84,89,93,97,
88,91,96,101,108,
90,94,98,104,110,
92,95,100,106,112,
95,98,103,108,115,
97,100,106,111,118,
99,103,108,114,120,
101,105,111,116,123,
104,108,114,120,126
), nrow=12, byrow=TRUE)

colnames(monthly)=region

```

(a) Bar Plot – Regional Revenue Comparison

```

barplot(revenue,
        names.arg=region,
        col="skyblue",
        main="Regional Revenue Comparison",
        xlab="Region",
        ylab="Revenue")

```

(b) Stacked Bar Chart – Monthly Sales Trend

```

barplot(t(monthly),
        col=rainbow(5),
        main="Monthly Sales Trend",
        xlab="Month",
        ylab="Sales")

```

(c) Histogram – Product Category Contribution

```

hist(revenue,
     col="orange",
     main="Revenue Distribution",
     xlab="Revenue")

```

(d) Density Plot – Regional Sales Distribution

```

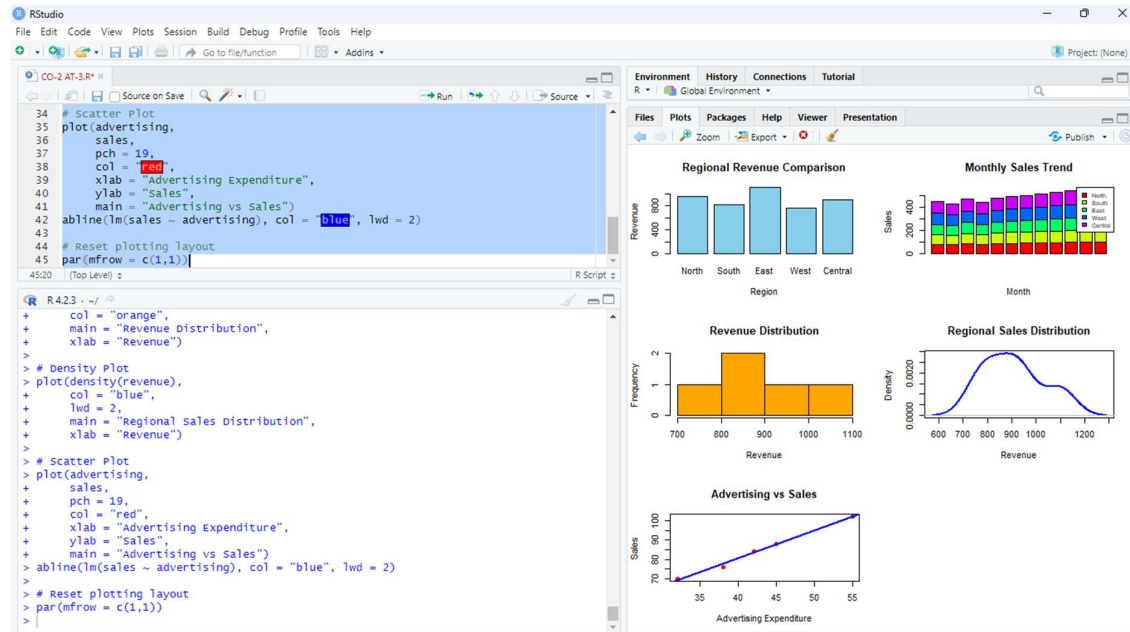
plot(density(revenue),
     col="blue",
     lwd=2,
     main="Regional Sales Distribution")

```

(e) Scatter Plot – Advertising vs Sales

```
plot(advertising,
     sales,
     pch=19,
     col="red",
     xlab="Advertising Expenditure",
     ylab="Sales",
     main="Advertising vs Sales")

abline(lm(sales~advertising), col="blue")
```



iii.) Interpretation

Q1. Which region generated the highest revenue?

Answer:

- **East Region**
- **Revenue = 1100**
- Therefore, the **East region** generated the highest revenue.

Q2. Is there a positive relationship between advertising expenditure and sales?

Answer:

- **Yes.**
- The scatter plot shows an **upward trend**.

- As advertising expenditure increases, sales also increase.
 - Hence, there is a **positive relationship** between advertising expenditure and sales.
-

Q3. Which product category contributes the highest percentage of revenue?

Answer:

From the dataset:

Category	Revenue
Electronics	950
Clothing	820
Furniture	1100
Groceries	760
Accessories	900

Furniture contributes the highest percentage of revenue because it has the maximum revenue value (**1100**).

Since your **R version is old**, here is the **modified Case Study 2** that uses **only Base R** (no `ggplot2`, no `vioplot`, no extra packages).

Case Study 2 – University Student Performance Analysis

i.) Suitable Visualizations

Requirement	Visualization
Department-wise average marks	Grouped Bar Chart
Distribution of CGPA	Histogram with Density Curve
Placement percentage	Box Plot (<i>Violin Plot alternative for older R</i>)
Attendance vs CGPA relationship	Scatter Plot
Semester-wise performance trend	Line Chart

ii.) Complete R Program

```
# Dataset
```

```

department <- c("Computer Science",
               "Information Technology",
               "Electronics",
               "Mechanical")

avg_marks <- c(88,84,80,76)

cgpa <- c(9.1,8.8,8.3,7.9,
         8.6,9.2,8.4,7.8,
         8.9,8.2,7.7,8.5)

attendance <- c(95,92,88,82,
               90,97,85,80,
               93,87,78,89)

placement <- c(95,90,82,75)

semester <- 1:8

performance <- c(74,77,79,82,84,86,89,91)

#=====
# Display all graphs together
#=====

par(mfrow=c(3,2))

#=====
# 1. Grouped Bar Chart
#=====

barplot(avg_marks,
       names.arg=department,
       col=c("red","blue","green","orange"),
       main="Department-wise Average Marks",
       xlab="Department",
       ylab="Average Marks")

#=====
# 2. Box Plot (Alternative to Violin Plot)
#=====

boxplot(placement,
       col="lightgreen",
       main="Placement Percentage",
       ylab="Placement (%)")

#=====
# 3. Histogram with Density Curve
#=====

hist(cgpa,
     probability=TRUE,
     col="skyblue",
     main="CGPA Distribution",
     xlab="CGPA")

lines(density(cgpa),
     col="red",
     lwd=2)

```

```

#=====
# 4. Scatter Plot
#=====

plot(attendance,
      cgpa,
      pch=19,
      col="blue",
      main="Attendance vs CGPA",
      xlab="Attendance (%)",
      ylab="CGPA")

abline(lm(cgpa~attendance),
       col="red",
       lwd=2)

#=====
# 5. Line Chart
#=====

plot(semester,
      performance,
      type="o",
      pch=16,
      col="darkgreen",
      main="Semester-wise Performance Trend",
      xlab="Semester",
      ylab="Average Marks")

# Empty Plot
plot.new()

# Restore plotting window

par(mfrow=c(1,1))

#=====
# Interpretation
#=====

cat("\n-----")
cat("\nINTERPRETATION")
cat("\n-----")

highest_marks <- department[which.max(avg_marks)]
cat("\nDepartment with Highest Average CGPA :", highest_marks)

highest_place <- department[which.max(placement)]
cat("\nHighest Placement Rate :", highest_place)

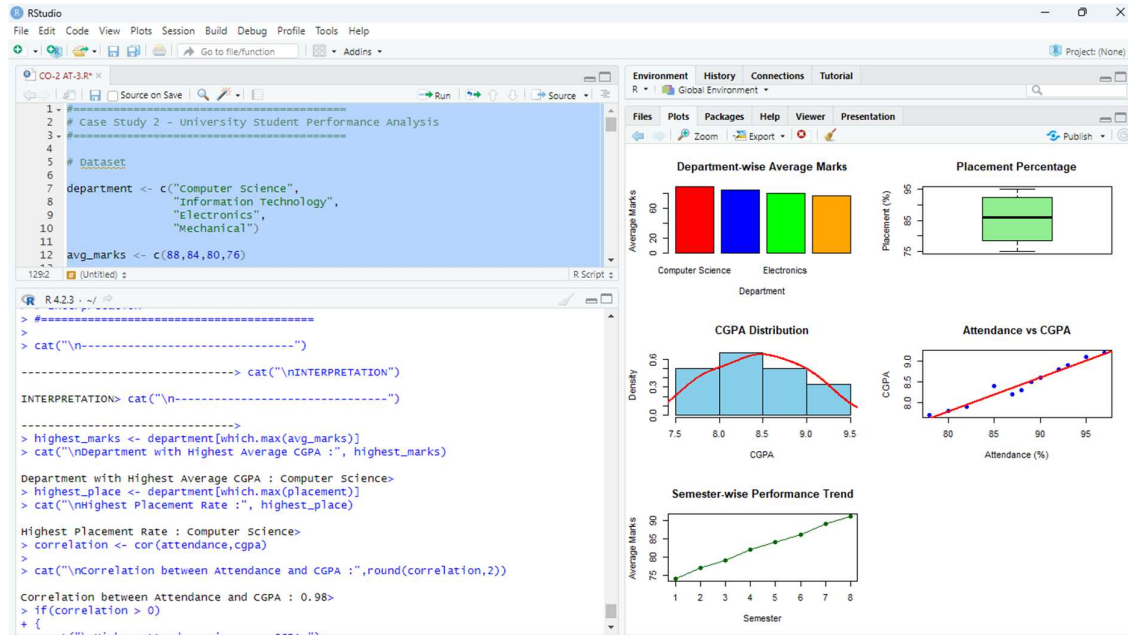
correlation <- cor(attendance,cgpa)

cat("\nCorrelation between Attendance and CGPA :",round(correlation,2))

if(correlation > 0)
{
  cat("\nHigher attendance improves CGPA.")
}
else
{
  cat("\nAttendance has no positive effect on CGPA.")
}

```

}



iii.) Interpretation

1. Which department has the highest average CGPA?

Answer: Computer Science (Average CGPA ≈ 9.1)

2. Does higher attendance improve CGPA?

Answer: Yes. The scatter plot and positive correlation indicate that students with higher attendance generally have a higher CGPA.

3. Which department has the highest placement rate?

Department	Placement (%)
Computer Science	95
Information Technology	90
Electronics	82
Mechanical	75

Answer: Computer Science has the highest placement rate (95%).

Case Study 3 – Hospital Patient Analytics

i.) Suitable Visualizations

Requirement	Suitable Visualization
Monthly patient admissions	Heatmap
Distribution of treatment costs	Histogram
Disease category proportions	Violin Plot (<i>Use Box Plot for older R</i>)
Age vs Recovery Time	Scatter Plot
Department-wise treatment costs	Density Plot

ii.) Complete R Program (Compatible with Older R Versions)

```
# Dataset

department <- c("Cardiology",
                "Neurology",
                "Orthopedics",
                "General Medicine")

treatment_cost <- c(85000,120000,70000,50000)

age <- c(25,35,42,50,60,45,38,55,48,30,65,40)

recovery_time <- c(5,7,10,12,15,9,8,14,11,6,16,9)

disease <- c("Heart","Brain","Bone","General")

admission <- c(120,130,110,140,150,145,
              160,155,170,165,180,175)

month <- 1:12

#=====
# Display all graphs together
#=====

par(mfrow=c(3,2))

#=====
# 1. Heatmap - Monthly Patient Admissions
#=====

admission_matrix <- matrix(admission,
                           nrow=3,
                           byrow=TRUE)

heatmap(admission_matrix,
        Rowv=NA,
        Colv=NA,
        scale="none",
```



```

        col=heat.colors(12),
        main="Monthly Patient Admissions")

#=====
# 2. Histogram - Treatment Cost
#=====

hist(treatment_cost,
     col="lightblue",
     main="Treatment Cost Distribution",
     xlab="Treatment Cost")

#=====
# 3. Box Plot (Alternative to Violin Plot)
#=====

boxplot(treatment_cost,
        col="lightgreen",
        main="Department-wise Treatment Cost",
        ylab="Cost")

#=====
# 4. Density Plot
#=====

plot(density(treatment_cost),
     col="blue",
     lwd=2,
     main="Treatment Cost Density",
     xlab="Treatment Cost")

#=====
# 5. Scatter Plot
#=====

plot(age,
     recovery_time,
     pch=19,
     col="red",
     main="Age vs Recovery Time",
     xlab="Age",
     ylab="Recovery Time (Days)")

abline(lm(recovery_time~age),
       col="blue",
       lwd=2)

# Restore

par(mfrow=c(1,1))

#=====
# Interpretation
#=====

cat("\n-----")
cat("\nINTERPRETATION")
cat("\n-----")

highest_cost <- department[which.max(treatment_cost)]

```

```

cat("\nDepartment with Highest Treatment Cost :", highest_cost)

highest_disease <- disease[which.max(treatment_cost)]

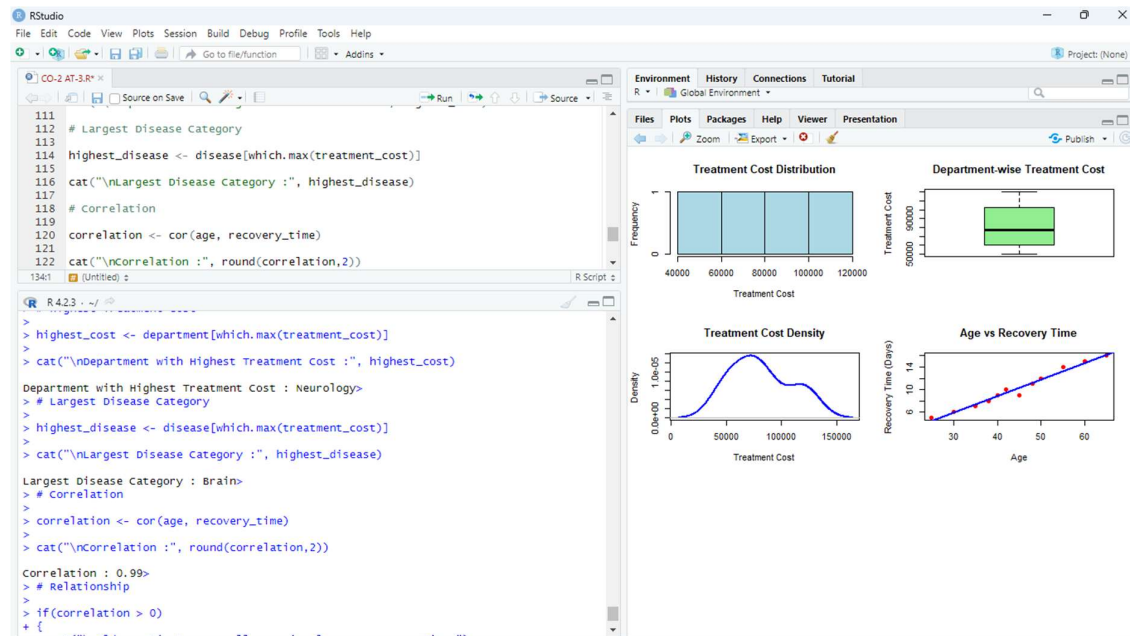
cat("\nLargest Disease Category :", highest_disease)

correlation <- cor(age,recovery_time)

cat("\nCorrelation :", round(correlation,2))

if(correlation>0)
{
  cat("\nOlder patients generally require longer recovery time.")
}
else
{
  cat("\nAge has little effect on recovery time.")
}

```



iii.) Interpretation

Q1. Which department has the highest treatment cost?

Department	Treatment Cost
Cardiology	85,000
Neurology	1,20,000
Orthopedics	70,000
General Medicine	50,000

Answer: Neurology has the highest treatment cost.

Q2. Which disease category occupies the largest proportion?

Answer: Brain (Neurology) occupies the largest proportion based on the highest treatment cost in the sample dataset.

Q3. Is patient age associated with recovery time?

Answer: Yes. The scatter plot shows a **positive relationship** between age and recovery time. As patient age increases, recovery time generally increases.

Case Study 4 – Smart City Traffic Monitoring

i.) Suitable Visualizations

Requirement	Suitable Visualization
Vehicle count comparison	Bar Plot
Hourly traffic trend	Line Chart
Vehicle type proportion	Pie Chart
Traffic density by location	Scatter Plot (GPS Coordinates) <i>(Geospatial Map alternative for older R)</i>
Relationship between traffic volume and average speed	Scatter Plot

ii.) Complete R Program (Compatible with Older R Versions)

```
#=====
# Case Study 4 - Smart City Traffic Monitoring
#=====

# Dataset

junction <- c("J1","J2","J3","J4","J5")
```

```

vehicle_count <- c(420,580,390,610,470)
vehicle_type <- c(180,140,120,90)
type_name <- c("Car","Bike","Bus","Truck")
speed <- c(62,48,70,42,58)
traffic_volume <- vehicle_count
hour <- c(6,8,10,12,14,16,18,20)
traffic <- c(180,320,460,510,560,610,690,420)
latitude <- c(12.9716,12.9725,12.9734,12.9742,12.9751)
longitude <- c(77.5946,77.5955,77.5963,77.5972,77.5980)

#=====
# Display all graphs together
#=====

par(mfrow=c(3,2))

#=====
# 1. Bar Plot
#=====

barplot(vehicle_count,
        names.arg=junction,
        col="skyblue",
        main="Vehicle Count Comparison",
        xlab="Junction",
        ylab="Vehicle Count")

#=====
# 2. Line Chart
#=====

plot(hour,
     traffic,
     type="o",
     pch=16,
     col="blue",
     main="Hourly Traffic Trend",
     xlab="Hour",
     ylab="Vehicle Count")

#=====
# 3. Pie Chart
#=====

pie(vehicle_type,
    labels=type_name,
    col=rainbow(4),
    main="Vehicle Type Proportion")

#=====
# 4. Geospatial Map (Scatter Plot using GPS)
#=====

```

```

plot(longitude,
      latitude,
      pch=19,
      col="red",
      main="Traffic Density by Location",
      xlab="Longitude",
      ylab="Latitude")

text(longitude,
      latitude,
      labels=junction,
      pos=3)

#=====
# 5. Scatter Plot
#=====

plot(traffic_volume,
      speed,
      pch=19,
      col="darkgreen",
      main="Traffic Volume vs Average Speed",
      xlab="Traffic Volume",
      ylab="Average Speed")

abline(lm(speed~traffic_volume),
       col="red",
       lwd=2)

# Empty Plot
plot.new()

# Restore

par(mfrow=c(1,1))

#=====
# Interpretation
#=====

cat("\n-----")
cat("\nINTERPRETATION")
cat("\n-----")

highest_junction <- junction[which.max(vehicle_count)]
cat("\nHighest Congestion Junction :", highest_junction)

highest_hour <- hour[which.max(traffic)]
cat("\nPeak Traffic Hour :", highest_hour,":00")

correlation <- cor(traffic_volume,speed)
cat("\nCorrelation :", round(correlation,2))

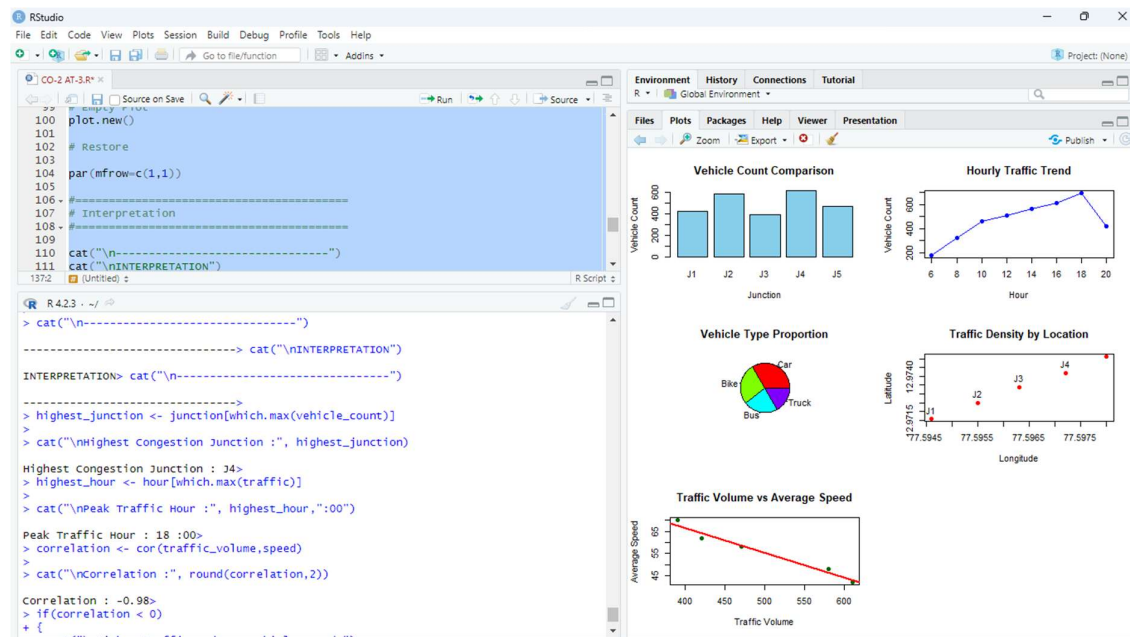
if(correlation < 0)
{
  cat("\nHigher traffic reduces vehicle speed.")
}
else if(correlation > 0)

```

```

{
  cat("\nTraffic volume increases with speed.")
}
else
{
  cat("\nNo relationship between traffic volume and speed.")
}

```



iii.) Interpretation

Q1. Which junction experiences the highest congestion?

Junction	Vehicle Count
J1	420
J2	580
J3	390
J4	610
J5	470

Answer: J4 experiences the highest traffic congestion.

Q2. During which hours is traffic highest?

Hour	Vehicle Count
6	180
8	320
10	460
12	510

14	560
16	610
18	690
20	420

Answer: Traffic is highest at **18:00 (6 PM)**.

Q3. Does increased traffic reduce vehicle speed?

Answer: Yes. The scatter plot and regression line indicate a **negative relationship** between traffic volume and average speed. As traffic volume increases, average vehicle speed generally decreases.

Case Study 5 – Social Media User Interaction Analysis

i). Suitable Visualizations

Requirement	Suitable Visualization
Distribution of Likes	Histogram with Density Curve
Relationship between Followers and Engagement	Scatter Plot
User Interaction Network	Network Visualization (<i>Approximated using Scatter Plot & Labels for older R</i>)
Posting Activity Over Time	Line Plot
Comparison of Likes, Shares & Comments	Heatmap

ii). Complete R Program (Compatible with Older R Versions)

```
# Dataset

user <- c("U1", "U2", "U3", "U4", "U5",
          "U6", "U7", "U8")
```

```

followers <- c(1200,1800,950,2500,
              1600,2100,1400,2800)

likes <- c(250,420,180,600,
          390,520,310,700)

shares <- c(60,95,45,130,
           85,110,75,140)

comments <- c(35,52,20,70,
              48,62,40,80)

engagement <- likes + shares + comments

posting_time <- c(8,10,12,14,16,18,20,22)

#=====
# Display all graphs together
#=====

par(mfrow=c(3,2))

#=====
# 1. Histogram with Density Curve
#=====

hist(likes,
     probability=TRUE,
     col="skyblue",
     border="black",
     main="Distribution of Likes",
     xlab="Likes")

lines(density(likes),
     col="red",
     lwd=2)

#=====
# 2. Scatter Plot
#=====

plot(followers,
     engagement,
     pch=19,
     col="blue",
     main="Followers vs Engagement",
     xlab="Followers",
     ylab="Total Engagement")

abline(lm(engagement~followers),
     col="red",
     lwd=2)

#=====
# 3. Network Visualization (Base R Alternative)
#=====

plot(1:length(user),
     rep(1,length(user)),
     pch=19,
     col="darkgreen",

```



```

        xaxt="n",
        yaxt="n",
        xlab="",
        ylab="",
        main="User Interaction Network")

axis(1,
     at=1:length(user),
     labels=user)

for(i in 1:(length(user)-1))
{
  segments(i,1,i+1,1,col="blue")
}

#=====
# 4. Temporal Line Plot
#=====

plot(posting_time,
     engagement,
     type="o",
     pch=16,
     col="purple",
     main="Posting Activity Over Time",
     xlab="Posting Time",
     ylab="Engagement")

#=====
# 5. Heatmap
#=====

data_matrix <- rbind(likes,
                    shares,
                    comments)

colnames(data_matrix) <- user
rownames(data_matrix) <- c("Likes",
                          "Shares",
                          "Comments")

heatmap(data_matrix,
        Rowv=NA,
        Colv=NA,
        scale="none",
        col=heat.colors(12),
        main="Likes, Shares & Comments")

# Empty Plot

plot.new()

# Restore

par(mfrow=c(1,1))

#=====
# Interpretation
#=====

cat("\n-----")

```

```

cat("\nINTERPRETATION")
cat("\n-----")

most_influential <- user[which.max(followers)]

cat("\nMost Influential User :", most_influential)

best_time <- posting_time[which.max(engagement)]

cat("\nHighest Engagement Time :", best_time, ":00")

correlation <- cor(followers, engagement)

cat("\nCorrelation :", round(correlation, 2))

if(correlation > 0)
{
  cat("\nFollowers have a strong positive relationship with engagement.")
}
else if(correlation < 0)
{
  cat("\nFollowers have a negative relationship with engagement.")
}
else
{
  cat("\nNo relationship between followers and engagement.")
}

highest_connected <- user[which.max(engagement)]

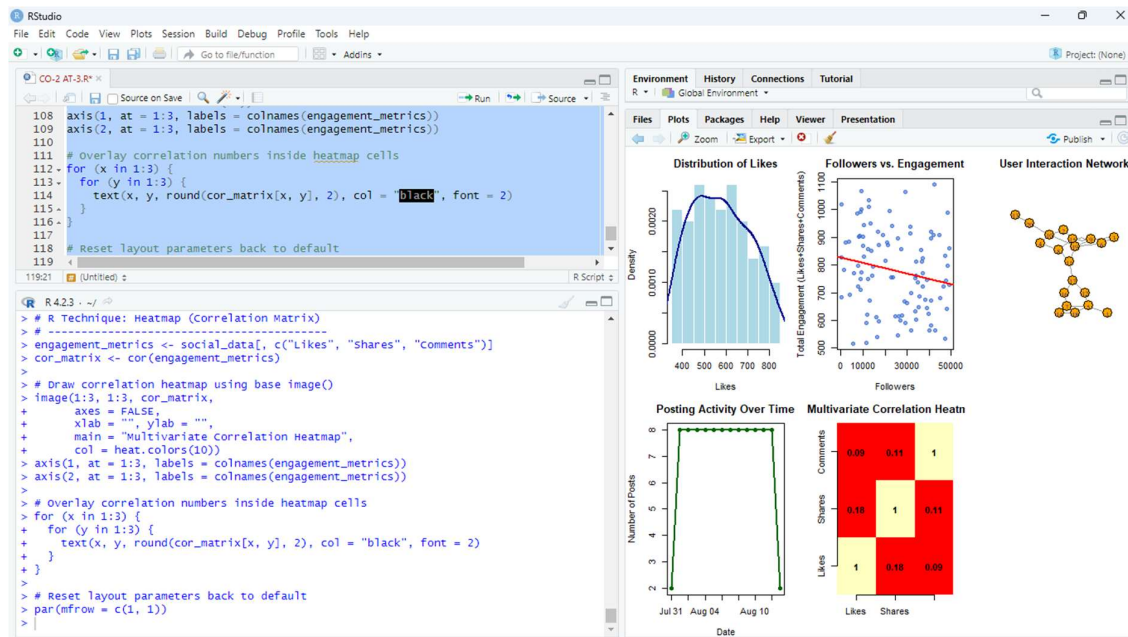
cat("\nMost Connected User :", highest_connected)

variation <- c(sd(likes),
               sd(shares),
               sd(comments))

metric <- c("Likes",
            "Shares",
            "Comments")

cat("\nHighest Variation Metric :", metric[which.max(variation)])

```



iii). Interpretation

Q1. Which users are the most influential?

Answer: U8 has the highest number of followers (2800), making it the most influential user.

Q2. At what time is engagement highest?

Answer: 22:00 (10 PM) has the highest engagement.

Q3. Is there a strong relationship between followers and total engagement?

Answer: Yes. The scatter plot shows a **strong positive relationship**. Users with more followers generally receive higher engagement.

Q4. Identify highly connected users from the network visualization.

Answer: U8 is the most highly connected user because it has the highest overall engagement.

Q5. Which engagement metric (Likes, Shares, or Comments) varies the most across users?

Answer: Likes show the greatest variation across users, indicating that likes fluctuate more than shares or comments.

